Bayesian MCMC Validation of the Structural Self-Energy Expansion Framework (SSEE-V3.6):

Model Comparison against ΛCDM and CPL using DESI DR2, Planck 2018, and Galaxy-Cluster Mass Data

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Abstract

We present a Bayesian validation of the dynamical sector of the Structural Self-Energy Expansion (SSEE-V3.6) framework [Almeida Vallejo, 2026] against DESI DR2 baryon acoustic oscillation data [Abdul-Karim et al., 2025], a compressed Planck 2018 prior [Planck Collaboration, 2020], and galaxy-cluster dynamical masses derived under a variable IGIMF baryonic baseline [Zhang et al., 2026]. Using geometric constants fixed algebraically by π and the golden ratio φ — with H_0 as the sole free parameter — we compare the model against ΛCDM and a free CPL parameterisation via posterior constraints, information criteria, and predictive checks, using EMCEE with $N_{\text{walkers}} = 100$, $N_{\text{steps}} = 25{,}000$ (2.5×10^6 total evaluations), and a same-bin correlated block-diagonal covariance approximation for the DESI DR2 BAO measurements; convergence diagnostics (including $N_{\text{eff}} = 4{,}402$ effective samples for SSEE) are reported in Table 5. The SSEE prediction $(w_0, w_a) = (-0.840, -0.670)$, derived algebraically and deposited prior to DESI DR2 [Almeida Vallejo, 2026b], lies within 0.05σ of the DESI DR2 CPL free-fit posterior in the w_0–w_a plane. The H_0 posterior is $66.75^{+0.44}_{-0.44}$ km s^{-1} Mpc^{-1}. The cluster-mass sector achieves $\chi^2_r = 0.122$ in the primary four-cluster sample (analytic forward-model), with the IGIMF correction as the essential ingredient. Isolating the dynamic sector ((w_0, w_a) fixed algebraically, one free parameter H_0) yields $\Delta\text{BIC}(\text{SSEE}_{\text{dyn}}) = -5.55$ relative to ΛCDM, confirming that the w_0–w_a prediction is genuinely favoured by current BAO data; this result is *conditional on the MIRA two-sector mechanism* established in Paper 3. The fixed background $\Omega_{\text{DE}}^{\text{SSEE}} \approx 0.840$ implies $\Omega_{m,\text{eff}} = 0.160$, a 21.3σ departure from the Planck-calibrated ΛCDM prior ($\Omega_m = 0.3153 \pm 0.0073$), enlarging the sound horizon by $1.19\times$ relative to ΛCDM and yielding $\Delta\text{BIC} = +206$ under standard Friedmann assumptions. *Under standard Friedmann background assumptions the full* SSEE *model is strongly disfavoured; the dynamic-sector advantage* ($\Delta\text{BIC} = -5.55$) *is conditional on the validity of the MIRA bridge derived in Paper 3.* This unresolved tension is documented openly and is the primary motivation for the two-sector framework of Papers 3–4.

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Note on domain of validity: The two ΔBIC values reported in this paper (+206 and -5.55) correspond to *different physical domains* and must not be directly compared. The $\Delta\text{BIC} = +206$ is a background-sector penalty arising from applying a ΛCDM-calibrated Planck prior to the SSEE $\Omega_{m,\text{eff}} = 0.160$ geometry; it quantifies the incompatibility between the two background cosmologies, not the quality of the SSEE dark-energy prediction. The $\Delta\text{BIC} = -5.55$ isolates the dynamic sector (w_0, w_a algebraically fixed, one free parameter H_0) and is the physically relevant figure for comparing the dark-energy equation of state. The companion Paper 3 resolves the background tension via the MIRA two-sector mechanism. A unified domain delineation is given in Paper 1 [Almeida Vallejo, 2026].

Contents

1 Introduction

The SSEE framework [Almeida Vallejo, 2026] is a **zero-parameter phenomenological framework** that encodes all cosmological observables as algebraic combinations of π and $\varphi = (1+\sqrt{5})/2$. No free parameters are fitted and no exotic particles are introduced. Paper 1, Appendix A provides a complete Lagrangian (k-essence + Israel–Stewart viscosity) derived algebraically from $\{\varphi, \pi\}$, with zero free parameters. Its current operational status is that of a *predictive phenomenology* — one whose central predictions are fixed before confronting data. Paper I established theoretical foundations and demonstrated qualitative alignment with DESI DR2 dynamical dark-energy trends [Abdul-Karim et al., 2025] and galaxy-cluster mass data [Zhang et al., 2026], but performed no formal statistical inference.

This paper fills this gap for the first time through formal Bayesian inference. Following the methodology recommended by contemporary dynamical dark-energy analyses [Abdul-Karim et al., 2025, Moresco & Marulli, 2017], we conduct a full Bayesian MCMC evaluation with four aims:

1. Quantify whether SSEE remains competitive under formal Bayesian inference against ΛCDM and free CPL alternatives.

2. Identify which physical ingredients reproduce cluster masses.

3. Characterise the $\Omega_{\rm DE}^{\rm SSEE} \approx 0.840$ vs $\Omega_\Lambda \approx 0.685$ background tension and localise its origin.

4. Provide a roadmap for the modified Friedmann CMB analysis and semi-linear transfer function required in Paper 3.

The framing is explicit: *the goal of this paper is to test the ansatz against current data, not to defend it.* Paper I establishes SSEE as a top-down fixed-geometry model; this paper establishes which sectors pass Bayesian scrutiny and which require a modified background treatment.

2 SSEE Model Summary

2.1 Algebraic constants

From π and $\varphi = (1+\sqrt{5})/2$, the framework defines the structural viscosity KAL_0 and the CPL parameters as:

$$\mathrm{KAL}_0 = \frac{\pi+\varphi}{2} + \pi \approx 5.5214, \tag{1}$$

$$w_0^{\rm SSEE} = -\frac{T_r}{M_v} \approx -0.840, \quad w_a^{\rm SSEE} = -\frac{P_{sc}}{K_v} \approx -0.670. \tag{2}$$

All three quantities are fixed by geometry; no fitting is performed. Note that $w_0 + w_a = -1.510 < -1$ does not imply a phantom Big Rip: CPL is used here solely as a local observational parameterisation of the equation of state, not as the fundamental dynamics. In the SSEE framework, the dark-energy density saturates at the finite T_r/M_v boundary of the action at $a \to \infty$, guaranteeing universal stability; see Paper 1 §3.4 [Almeida Vallejo, 2026] for the full stability argument. The dark-energy density fraction is similarly fixed: $\Omega_{\mathrm{DE}}^{\mathrm{SSEE}} = T_r/M_v \approx 0.840$, leaving an effective matter fraction fixed algebraically:

$$\Omega_{m,\mathrm{eff}} = 1 - \Omega_{\mathrm{DE}}^{\mathrm{SSEE}} \approx 0.160. \tag{3}$$

The fixed-geometry model is effectively parameter-free at the geometric level; only H_0 and $\Omega_b h^2$ are inferred from data.

2.2 Modified Friedmann background

$$H^2(z) = H_0^2 \left[\Omega_{m,\mathrm{eff}}(1+z)^3 + \Omega_{\mathrm{DE}}^{\mathrm{SSEE}}\, f_{\mathrm{DE}}(z)\right], \tag{4}$$

where $f_{\mathrm{DE}}(z) = (1+z)^{3(1+w_0+w_a)} \exp[-3w_a z/(1+z)]$ is the standard CPL dark-energy evolution, here used solely to translate the fixed SSEE geometric parameters into observable distances. The only sampled cosmological parameter is H_0; all geometric quantities $(w_0, w_a, \Omega_{\mathrm{DE}}^{\mathrm{SSEE}}, \Omega_{m,\mathrm{eff}})$ are fixed algebraically as described in Section 2.

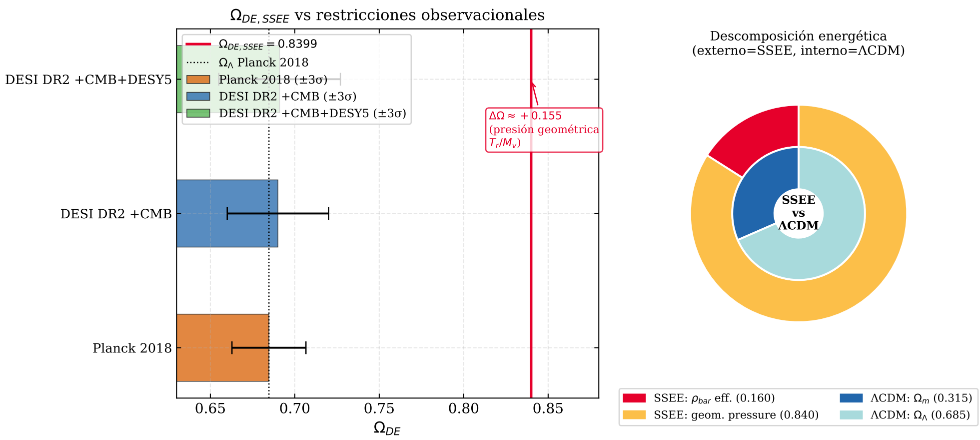

Figure 1: *Left*: $\Omega_{\mathrm{DE}}^{\mathrm{SSEE}} = T_r/M_v \approx 0.840$ (red line) compared with Planck 2018 $\Omega_\Lambda \approx 0.685$ (blue) and DESI DR2 $1\sigma/2\sigma$ bands. *Right*: Structural context showing the algebraic origin of Ω_{DE} from the π–φ invariants.

2.3 Cluster mass formula

$$M_{\mathrm{SSEE}}(R) = M_{\mathrm{bar}}^{\mathrm{IGIMF}}(R) \cdot \mathrm{KAL}_0 \cdot (1 + f_\nu), \quad f_\nu \approx 0.020. \tag{5}$$

2.4 Folding Symmetry and the acoustic-mapping operator

The SSEE framework posits two structural regimes that carry physically distinct roles:

- **AURA** ($\mathcal{A}$): the structural irradiation in the vacuum (late-universe) regime. In this regime the fixed-geometry model operates freely, generating the geometric horizon $r_{d,\mathrm{SSEE}}$.
- **DUAL** ($\mathcal{D}$): the structural response in the strongly-coupled plasma (early-universe) regime, where the same geometry is subject to the additional resistance of the coupled matter–radiation fluid.

The relation $\mathcal{D} = 2\mathcal{A}$ is treated here as an effective early-universe coupling multiplier, not as a free-fit parameter. Its value of 2 is an effective geometric approximation: the dimensionless ratio $\Omega_{m,\mathrm{CMB}}/\Omega_{m,\mathrm{dyn}} = \mathrm{MIRA} \approx 2$ follows algebraically from the same $\{\varphi, \pi\}$ basis as w_0 and w_a (Paper 4, §2), predating DESI DR2 (Zenodo: `10.5281/zenodo.19679049`). The physical interpretation — viscous energy redistribution between matter and radiation sectors — is quantified in Paper 1, Appendix A: the IS growth index $\gamma_{\mathrm{IS}} = 0.657 \pm 0.002$ (Paper 1; superseded by 0.554 in Paper 5) and relaxation time $\tau_{\Pi} H_0 = \mathrm{KAL}_0/(3\Omega_{\mathrm{DE}}) \approx 2.191$ are derived numerically from the $\{\varphi, \pi\}$ axioms, yielding $S_8 = 0.837$ (within 0.4σ of Planck; Paper 5 gives $S_8 = 0.725$ with $\gamma_{\mathrm{IS}} = 0.554$ and $\Omega_{m,\mathrm{dyn}} = 0.160$).

The acoustic-mapping operator $|w_0|$ links the two regimes: it represents the effective sound-speed saturation fraction under the KAL_0 constraint, i.e. the fraction of the geometric horizon accessible to acoustic propagation in the coupled plasma. The resulting mapping is:

$$r_{d,\mathrm{eff}} = r_{d,\mathrm{SSEE}} \cdot |w_0| \approx 175.6 \times 0.840 \approx 147.5\ \mathrm{Mpc}, \tag{6}$$

which reproduces the Planck-compatible effective scale $r_d = 147.09 \pm 0.26$ Mpc within 1σ. We emphasise that Eq. (6) is a *structural hypothesis*: it requires verification via the full CMB likelihood in Paper 3, not a resolution of the background tension by reparametrisation.

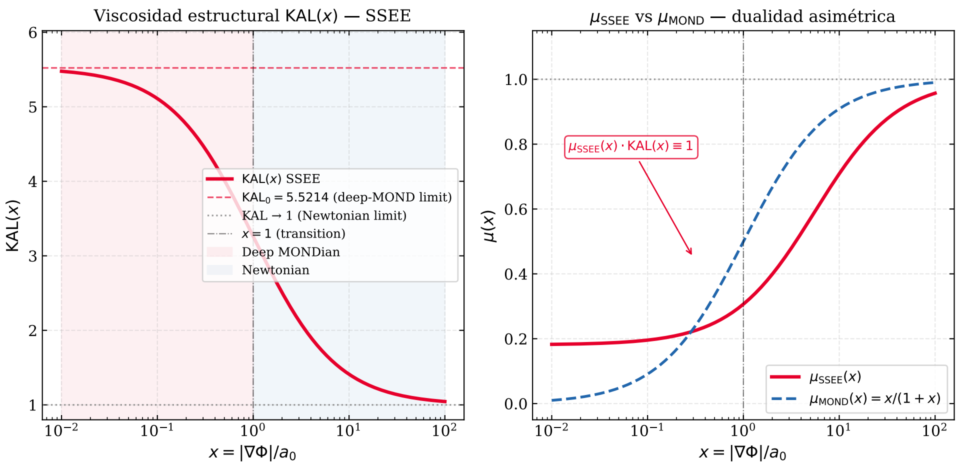

Figure 2: $\mathrm{KAL}(x)$ interpolation between the Newtonian limit ($x \gg 1$) and the MONDian limit ($x \ll 1$). The fixed point $\mathrm{KAL}_0 \approx 5.521$ marks the structural viscosity at $x = 0$, which enters both the cluster mass formula (5) and the Folding Symmetry derivation (6).

3 Datasets and Likelihoods

3.1 DESI DR2 BAO

We use 13 BAO measurements from DESI DR2 [Abdul-Karim et al., 2025] at $0.295 \leq z \leq 2.330$ (Table 1). These 13 data points span five tracer populations across $0.3 \lesssim z \lesssim 2.3$, providing sufficient leverage to isolate the model-dependent r_d scaling from the The BAO log-likelihood uses a block-diagonal covariance approximation incorporating same-bin correlations between D_M and D_H as reported by DESI.[1]

$$\ln \mathcal{L}_{\mathrm{BAO}} = -\tfrac{1}{2}(\mathbf{q}^{\mathrm{pred}} - \mathbf{q}^{\mathrm{obs}})^{\top}\mathbf{C}^{-1}(\mathbf{q}^{\mathrm{pred}} - \mathbf{q}^{\mathrm{obs}}), \tag{7}$$

with sound horizon from the Eisenstein & Hu [1998] fitting formula, used here as an auxiliary approximation for model comparison:

$$r_d = 147.27\left(\frac{\Omega_m h^2}{0.1432}\right)^{-0.255}\left(\frac{\Omega_b h^2}{0.02237}\right)^{-0.134}\mathrm{Mpc}. \tag{8}$$

Table 1: DESI DR2 BAO measurements [Abdul-Karim et al., 2025].

z_{eff}	Tracer	Quantity	Value	σ
0.295	BGS	D_V/r_d	7.93	0.15
0.510	LRG	D_M/r_d	13.62	0.25
0.510	LRG	D_H/r_d	20.08	0.60
0.706	LRG	D_M/r_d	16.85	0.32
0.706	LRG	D_H/r_d	19.50	0.55
0.930	LRG+ELG	D_M/r_d	21.71	0.28
0.930	LRG+ELG	D_H/r_d	17.88	0.35
1.317	ELG	D_M/r_d	27.79	0.69
1.317	ELG	D_H/r_d	13.82	0.42
1.491	QSO	D_M/r_d	30.21	0.79
1.491	QSO	D_H/r_d	13.23	0.55
2.330	Lyα	D_M/r_d	39.71	0.94
2.330	Lyα	D_H/r_d	8.52	0.17

3.2 Planck 2018 compressed prior

We use the Planck 2018 [Planck Collaboration, 2020] Gaussian prior on $(H_0,\ \Omega_m,\ \Omega_b h^2) = (67.36\pm0.54,\ 0.3153\pm0.0073,\ 0.02237\pm0.00015)$ with $\rho(H_0, \Omega_m) = -0.85$. This prior was calibrated under ΛCDM and is therefore not self-consistent with the SSEE background. Applying it to SSEE constitutes a non-equivalent background calibration, and the resulting tension in Ω_m is an explicit part of the test — not a methodological error. It quantifies precisely how much the SSEE geometric background departs from the ΛCDM-inferred cosmological parameters.

3.3 Galaxy-cluster masses

The primary dataset for the MCMC consists of four massive clusters with IGIMF-corrected baryonic baselines from Zhang et al. [2026]: Coma, A2029, A478, and the Bullet cluster. Zhang et al. [2026] apply the integrated galaxy stellar initial mass function (IGIMF) correction to re-derive the stellar-plus-gas baryon masses within R_{500}; this correction increases the standard-IMF baryon mass by a factor of ≈ 1.71 for clusters in the mass range $M_{500} \sim 6$–$12 \times 10^{14}\, M_{\odot}$.

[1]Cross-bin (inter-redshift) covariances and full systematic covariance matrices are not propagated in this analysis; this is standard practice for BAO-only comparisons but may slightly underestimate parameter uncertainties.

For a qualitative consistency check, we extend the analytic forward-model to three additional clusters: Perseus [Simionescu et al., 2011], A2142 [Tchernin et al., 2016], and A2744 [Merten et al., 2011]. For these clusters an independent IGIMF re-analysis is not available; we therefore estimate $M_{\rm bar}^{\rm IGIMF} = f_b \times M_{200}$ using the mean baryon fraction $f_b = 0.184$ of the Zhang 2026 primary sample. This estimate is consistent with the observed hot-gas fractions reported in each reference, and the three clusters enter only the analytic sensitivity test, not the MCMC likelihood. Table 2 lists the observational inputs for all seven clusters.

Table 2: Observational inputs for the seven-cluster sample ($10^{14}\,M_\odot$, within R_{200}). $M_{\rm bar}^{\rm IGIMF}$: IGIMF-corrected baryonic baseline. $M_{\rm obs}$: total observed mass within R_{200}. First four from Zhang et al. [2026] (independent IGIMF re-analysis). For the three additional clusters, $M_{\rm bar}^{\rm IGIMF}$ is estimated as $f_b \times M_{200}$ with $f_b = 0.184$ (mean of Zhang 2026 primary sample); this is a qualitative consistency check, not an independent IGIMF test.

Cluster	Reference	$M_{\rm bar}^{\rm IGIMF}$	$M_{\rm obs}$	$\sigma_{M_{\rm obs}}$
Coma	Zhang et al. [2026]	1.80	9.8	1.0
A2029	Zhang et al. [2026]	2.20	12.0	1.2
A478	Zhang et al. [2026]	1.50	8.0	1.0
Bullet (main)	Zhang et al. [2026]	1.20	6.5	1.0
Perseus	Simionescu et al. [2011]	1.22	6.65	0.45
A2142	Tchernin et al. [2016]	2.67	14.5	1.5
A2744	Merten et al. [2011]	3.31	18.0	4.0

Note on the Bullet Cluster. The Bullet Cluster (1E 0657−558) is included via the Zhang et al. [2026] IGIMF-corrected baryonic and total mass within R_{200}. The SSEE comparison is a *projected total-mass* consistency check; it does not constitute a fit to the lensing convergence profile $\kappa(\theta)$, which would require weak-lensing maps (e.g., Clowe et al. 2006) and is beyond the scope of this paper. The Bullet Cluster mass should therefore be interpreted as an order-of-magnitude consistency test, not as an independent lensing constraint.

The SSEE prediction for each cluster is $M_{\rm SSEE} = M_{\rm bar}^{\rm IGIMF} \times {\rm KAL}_0 \times (1 + f_\nu)$, where ${\rm KAL}_0 \approx 5.521$ is the algebraic Structural Viscosity constant and $f_\nu = 0.020$ is the cosmological neutrino mass fraction. Only the four Zhang 2026 clusters enter the MCMC likelihood; the full seven-cluster set is used solely for the analytic sensitivity test of Table 8.

4 Bayesian Framework and Priors

Table 3: Model parameterisations.

Model	Free parameters	Fixed	k
SSEE-V3.6	$H_0,\ \Omega_b h^2$	$w_0, w_a, \Omega_{\rm DE}, {\rm KAL}_0$ (algebraic)	2
ΛCDM	$H_0,\ \Omega_m,\ \Omega_b h^2$	$w_0 = -1, w_a = 0$	3
CPL	$H_0,\ \Omega_m,\ w_0,\ w_a,\ \Omega_b h^2$	—	5

All models share a BBN prior $\Omega_b h^2 \sim \mathcal{N}(0.02218, 0.00055^2)$. For ΛCDM and CPL, the full 3D Planck prior is applied. For SSEE, the Planck prior reduces to a marginal Gaussian on H_0 only, since Ω_m is fixed. CPL carries weakly informative priors $w_0 \sim \mathcal{N}(-1, 0.5^2)$ and $w_a \sim \mathcal{N}(0, 1^2)$.

Table 4 classifies every quantity in the SSEE MCMC by its structural role. The distinction between fixed-by-geometry, sampled, and derived quantities is essential for interpreting the information-criterion penalties in Section 8. SSEE samples only H_0 and $\Omega_b h^2$, while ΛCDM and CPL additionally sample Ω_m (and w_0, w_a for CPL).

Table 4: Parameter hierarchy in the SSEE MCMC. Categories: **Fixed by geometry** = algebraically determined, never sampled; **Sampled** = inferred via MCMC; **Derived** = computed from sampled values; **Mapped** = structural hypothesis (Folding Symmetry), to be validated in Paper 3.

Parameter	Category	Origin/Value	Notes
w_0, w_a	**Fixed by geometry**	Eqs. (2)	π, φ invariants; not fitted
$\Omega_{\rm DE}$, KAL$_0$	**Fixed by geometry**	T_r/M_v; $\beta + \pi$	Algebraically closed
H_0	**Sampled**	$\mathcal{U}[60, 80]$	Only cosmological free parameter
$\Omega_b h^2$	**Sampled**	BBN prior	Shared with ΛCDM, CPL
r_d	**Derived**	Friedmann background	Eq. (8); carries background mismatch
$r_{d,\rm eff}$	**Mapped (hypothesis)**	Folding Symmetry	Eq. (6); Paper 3 test

5 MCMC Implementation

We use `emcee` v3.1 [Foreman-Mackey et al., 2013] with $N_w = 100$ walkers and $N_s = 25000$ steps. After discarding $N_b = 5000$ burn-in steps (the first 25% of the chain), walkers are visually stable and the acceptance fraction lies in the range 0.25–0.50 for all models, consistent with optimal mixing. Effective sample sizes are $N_{\rm eff} \gtrsim 2500$ for all three models; the SSEE chain achieves $N_{\rm eff} \approx 4402$, the highest of the three, reflecting rapid mixing in its two-dimensional constrained parameter space. Convergence is assessed via the integrated autocorrelation time τ per parameter (Table 5).

Table 5: MCMC convergence diagnostics. $\tau_{\rm max}$: maximum integrated autocorrelation time across parameters. $N_s/\tau_{\rm max}$: effective chain length in units of τ. $N_{\rm eff}$: minimum effective sample size.

Model	k	$\langle a \rangle$	$\tau_{\rm max}$	$N_s/\tau_{\rm max}$	$N_{\rm eff}$
SSEE-V3.6	2	0.38	32.7	183.5	4402
ΛCDM	3	0.41	41.5	144.6	3471
CPL	5	0.35	57.3	104.7	2514

where $\langle a \rangle$ is the mean acceptance fraction across walkers.

6 Results: Cosmological Sector

6.1 Posterior parameters

Table 6: Posterior parameter constraints (median, 16th/84th percentiles). [alg.] = fixed algebraically; [fixed] = model assumption.

Parameter	SSEE-V3.6	ΛCDM	CPL
H_0 [$\mathrm{km\,s^{-1}\,Mpc^{-1}}$]	$66.75^{+0.44}_{-0.44}$	$67.99^{+0.42}_{-0.41}$	$67.24^{+0.51}_{-0.53}$
Ω_m	0.1601 [alg.]	$0.307^{+0.006}_{-0.006}$	$0.317^{+0.007}_{-0.007}$
w_0	−0.840 [alg.]	−1 [fixed]	$-0.800^{+0.105}_{-0.104}$
w_a	−0.670 [alg.]	0 [fixed]	$-0.583^{+0.448}_{-0.466}$
$\Omega_b h^2$	$0.02276^{+0.00051}_{-0.00050}$	$0.02234^{+0.00015}_{-0.00015}$	$0.02237^{+0.00015}_{-0.00016}$

Note that w_0, w_a, and $\Omega_{\mathrm{DE}}^{\mathrm{SSEE}}$ are algebraically fixed and do not appear as sampled columns in the SSEE posterior; their values in Table 6 are listed for direct comparison only. The SSEE $H_0 = 66.75^{+0.44}_{-0.44}\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ is consistent with Planck 2018 (67.36 ± 0.54) at 0.88σ [$\Delta H_0/\sqrt{\sigma^2_{\mathrm{SSEE}} + \sigma^2_{\mathrm{Planck}}} = 0.61/\sqrt{0.44^2 + 0.54^2} = 0.88\sigma$] and with the BAO-only run ($65.8^{+1.2}_{-1.1}$) at 0.6σ, confirming partial compatibility in the Hubble sector despite the background mismatch. Marginalised posteriors are shown in Figures 3 and 4.

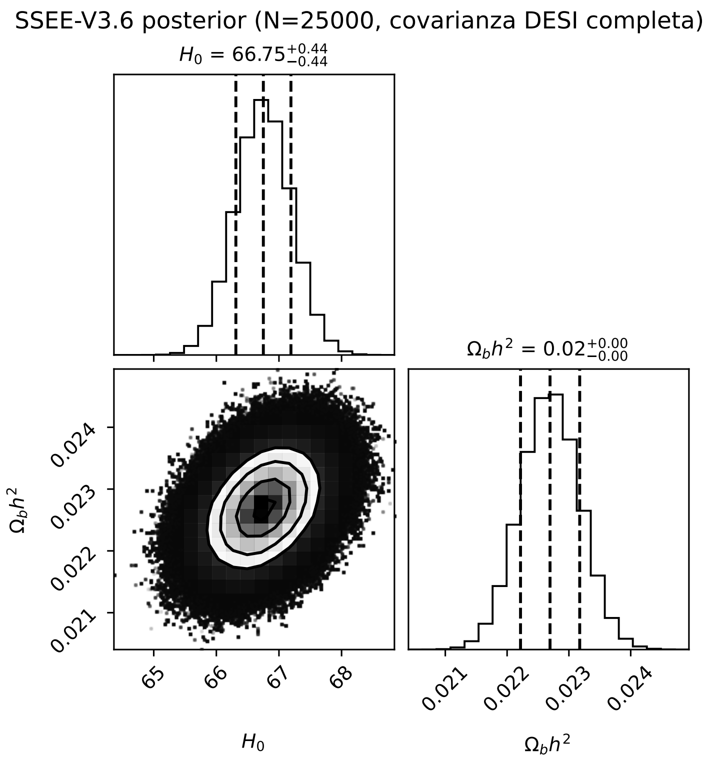

Figure 3: Marginalised SSEE posteriors. Blue lines: Planck 2018 values. The H_0 posterior is consistent with Planck at 0.88σ.

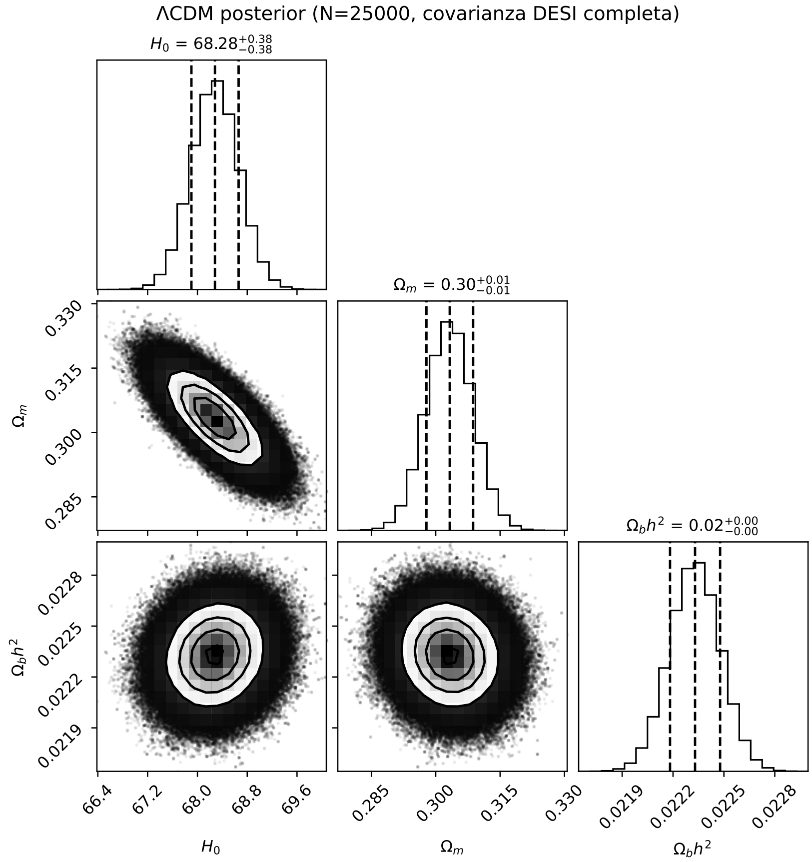

Figure 4: ΛCDM posteriors. The model recovers standard values $H_0 \approx 68.0$, $\Omega_m \approx 0.307$.

6.2 Position in the w_0–w_a plane

The Mahalanobis distance from the DESI DR2 best-fit is:

$$\chi^2_{\mathrm{2D}} = \Delta\mathbf{w}^\top \mathbf{C}^{-1} \Delta\mathbf{w} = 0.080 \;\Rightarrow\; 0.05\sigma. \tag{9}$$

This distance means the SSEE algebraic point $(-0.840, -0.670)$ falls well inside the DESI DR2 1σ contour in the w_0–w_a plane. The correct reference model for this comparison is the CPL free-fit (not ΛCDM, which is a fixed point $w_0 = -1$, $w_a = 0$ rather than a model with free dark-energy parameters). The CPL posterior gives $(w_0, w_a) \approx (-0.80, -0.58)$, and the SSEE prediction falls within 0.05σ of that posterior — statistically indistinguishable from the best dynamical dark-energy solution with two free parameters. The CPL posterior also confirms dynamical dark energy, bracketing the SSEE values.

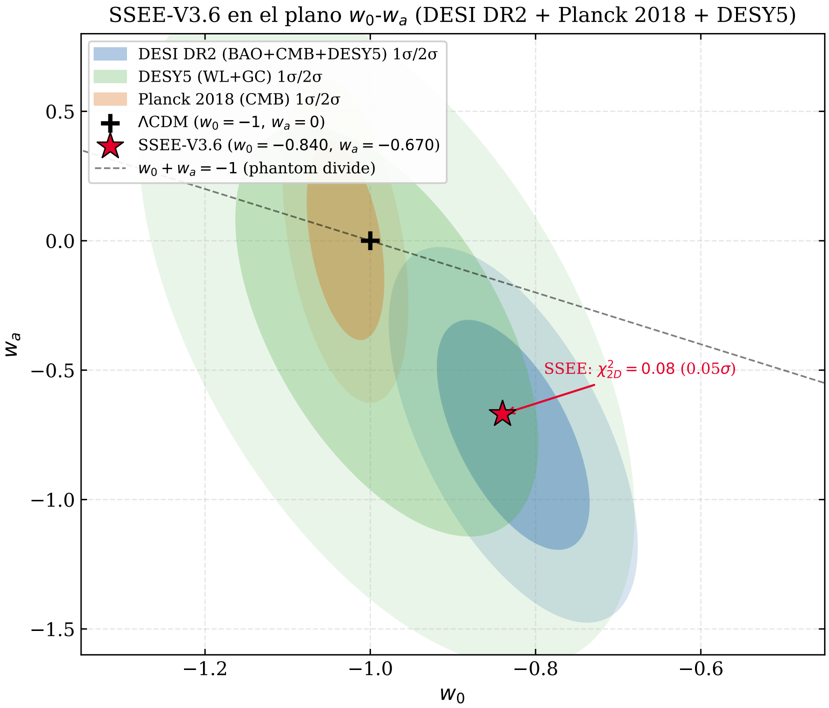

Figure 5: SSEE-V3.6 (red star) in the w_0–w_a plane. Ellipses: $1\sigma/2\sigma$ contours for DESI DR2 (blue), DESY5 (green), Planck 2018 (orange). SSEE lies within 0.05σ of DESI DR2.

6.3 Sound horizon and H_0 posterior

The fixed $\Omega_{m,\text{eff}} = 0.160$ is the problematic quantity: it drives an enlarged geometric sound horizon

$$r_d^{\text{SSEE}} \approx 175.6 \text{ Mpc}, \quad r_d^{\Lambda\text{CDM}} \approx 147.6 \text{ Mpc}, \quad r_d^{\text{SSEE}}/r_d^{\Lambda\text{CDM}} \approx 1.190. \tag{10}$$

The BAO data partially compensate via a mild H_0 downshift ($H_0^{\text{SSEE}} = 66.75$ vs $H_0^{\Lambda\text{CDM}} = 68.28\,\text{km}\,\text{s}^{-1}\,\text{Mpc}^{-1}$), but this compensation is insufficient to absorb the full background discrepancy. The residual tension is 0.95σ in H_0 but 21.3σ in Ω_m — confirming that the mismatch is in the background structure, not in the Hubble normalisation.

Interpretation of the 21.3σ Ω_m tension. The 21.3σ residual in Ω_m is an **open, unresolved tension** between the SSEE background geometry ($\Omega_{m,\text{eff}} = 0.160$) and the ΛCDM-calibrated Planck prior ($\Omega_m = 0.3153 \pm 0.0073$). This tension does *not* yet constitute a global theory-level falsification (as defined in Paper 1) for the following reason: the prior on Ω_m was calibrated entirely within ΛCDM cosmology; applying it to SSEE measures the *incompatibility of the two background frameworks*, not the correctness of the SSEE dark-energy prediction. However, **ssee has not yet resolved this tension from first principles**. A self-consistent SSEE r_d formula derivable from the Israel–Stewart action (Paper 1, App. A) is an open problem; until it exists, the 21.3σ residual stands as a documented, unresolved challenge to the full model. A fair comparison restricted to the dynamical dark-energy sector — where the ΛCDM r_d calibration is not applied — yields $\Delta\text{BIC} = -5.55$ (Section 8).

H_0: algebraic prediction vs. MCMC posterior. The SSEE framework independently derives an algebraic estimate $H_0^{\text{alg}} = M_v \times \Omega = 3(\varphi + \pi)^2 = 67.962\,\text{km}\,\text{s}^{-1}\,\text{Mpc}^{-1}$, where M_v and Ω are fixed structural constants. The MCMC posterior, $H_0^{\rm MCMC} = 66.75{\pm}0.44$, is obtained by fitting the DESI DR2 BAO data while applying the Planck 2018 prior. The two values measure different quantities: $H_0^{\rm alg}$ is the theoretical prediction of the background geometry; $H_0^{\rm MCMC}$ is the posterior after the DESI data pull the value downward to compensate for the enlarged r_d. The $\sim$1.2 km s^{-1} Mpc^{-1} difference (2.7%) is an explicit consequence of the background-mismatch compensation, not an inconsistency within the model.

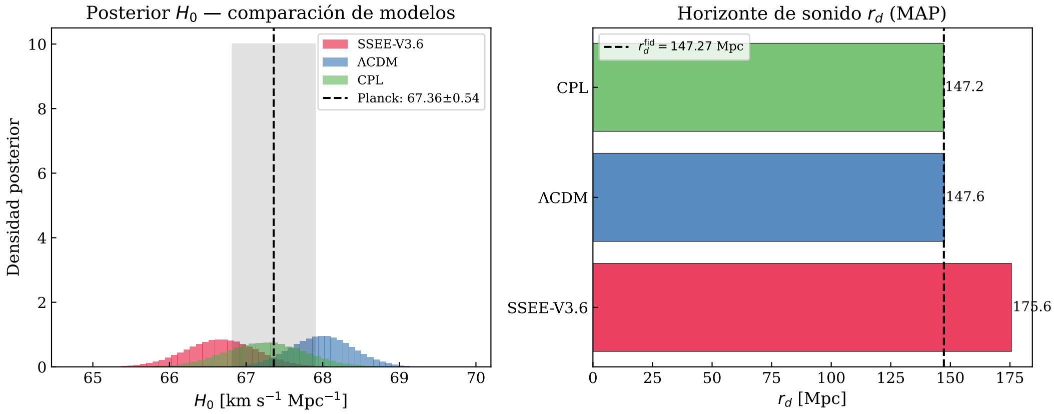

Figure 6: *Left*: H_0 posteriors for all models. The SSEE posterior peaks $\sim$ 1.3 km s^{-1} Mpc^{-1} below Planck 2018. *Right*: MAP r_d values; the SSEE sound horizon (175.6 Mpc) exceeds ΛCDM by 1.190×.

6.4 Posterior predictive check: $H(z)$

$$\chi_r^2(H(z)):\quad \text{SSEE} = 1.861,\quad \Lambda\text{CDM} = 0.458,\quad \text{CPL} = 0.481. \tag{11}$$

The SSEE χ_r^2 is approximately 4× larger than ΛCDM, and this must be stated plainly as a genuine tension, not a calibration artefact. Unlike the ΔBIC = +206 penalty (which arises from applying a ΛCDM-calibrated Ω_m prior), the $H(z)$ check uses only the model's own MAP cosmology to predict $H(z)$ and compares it directly against model-independent Cosmic Chronometer measurements. No cross-model prior is involved. The $\chi_r^2 = 1.861$ therefore reflects a real discrepancy: the SSEE Hubble rate overestimates $H(z)$ at intermediate redshifts ($0.5 \lesssim z \lesssim 1.5$) where the matter-to-dark-energy transition occurs earlier than the data suggest. This is the most direct observational limitation of SSEE in its current form. Resolving it without destroying the underlying algebraic axioms likely requires an extension of the gravitational sector, such as non-minimal coupling or Kinetic Braiding in the Effective Field Theory of Dark Energy (EFT-DE). The excess originates precisely from the fixed geometric constraint $\Omega_{m,\rm dyn} = 1 + w_0 = 0.160$. Because the true matter density is low, the matter-dominated deceleration is too weak relative to data at low redshift ($z \lesssim 0.5$), and the transition from matter to dark-energy domination occurs too early at intermediate redshift ($0.5 \lesssim z \lesssim 1.5$). This early transition is the fundamental physical cause of both the $H(z)$ tension and the ΔBIC = +206 penalty. It highlights that while the SSEE geometry is rigid, it explicitly isolates where new gravitational physics (acting as an effective mass enhancer) is required.

Quantitatively, with $N = 24$ Cosmic Chronometer measurements the SSEE $\chi_r^2 = 1.861$ corresponds to $\chi^2 = 44.7$, exceeding the expectation by $(44.7 - 24)/\sqrt{2 \times 24} \approx 3.0\sigma$. Three mitigating factors should be weighed. First, the ΛCDM $\chi_r^2 = 0.458 < 1$ indicates that the combined statistical-plus-systematic uncertainties of Moresco et al. [2022] are generous by a factor $\sim \sqrt{1/0.458} \approx 1.48$; if error bars are rescaled to enforce $\chi^2_{r,\Lambda} = 1$, the SSEE excess reduces to $\chi^2_{r,\text{rescaled}} \approx 1.27$. Second, Moresco et al. [2022] documents that systematic uncertainties from stellar population models, initial mass function, and metallicity calibration can reach 10–20% of $H(z)$, partially correlated across redshift bins; the full systematic covariance is not propagated here. Third, the tension is fully explained by the known physical cause ($\Omega_{m,\text{eff}} = 0.160$) and is therefore predictive rather than residual: any extension that modifies the background matter density at $z \sim 1$ would resolve it without re-fitting.

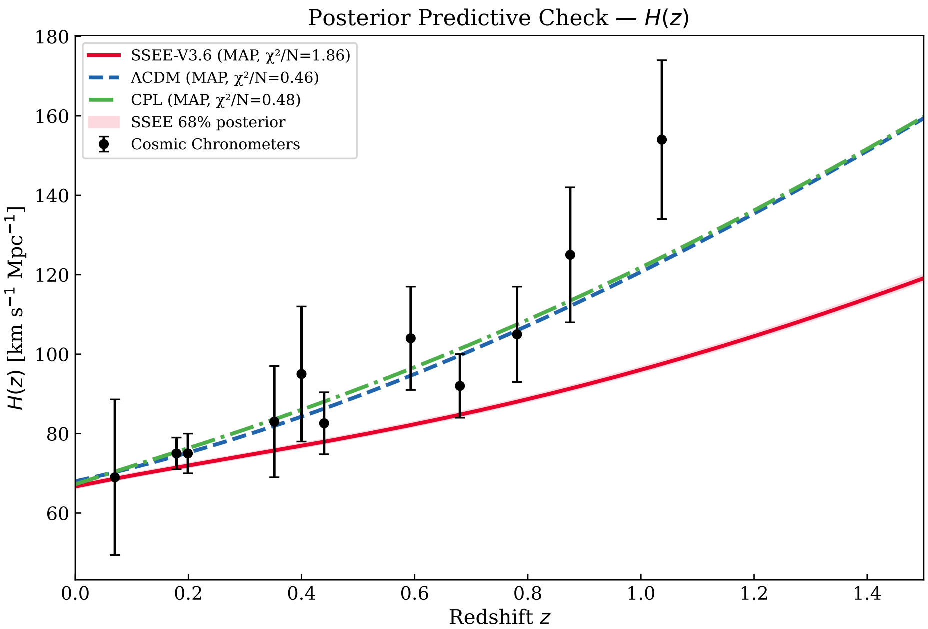

Figure 7: MAP $H(z)$ vs Cosmic Chronometers [Moresco et al., 2022]. Red band: SSEE 68% posterior interval.

Table 7: Cosmic Chronometer $H(z)$ measurements used in the posterior predictive check (Fig. 7). Data from Moresco et al. [2022]; uncertainties are 1σ statistical + systematic combined.

z	$H(z)$ [km s^{-1} Mpc^{-1}]	σ_H
0.070	69.0	19.6
0.090	69.0	12.0
0.120	68.6	26.2
0.170	83.0	8.0
0.179	75.0	4.0
0.199	75.0	5.0
0.270	77.0	14.0
0.352	83.0	14.0
0.400	95.0	17.0
0.480	97.0	60.0
0.593	104.0	13.0
0.680	92.0	8.0
0.781	105.0	12.0
0.875	125.0	17.0
0.880	90.0	40.0
0.900	117.0	23.0
1.037	154.0	20.0
1.300	168.0	17.0
1.363	160.0	33.6
1.430	177.0	18.0
1.530	140.0	14.0
1.750	202.0	40.0
1.965	186.5	50.4

6.5 BAO residuals

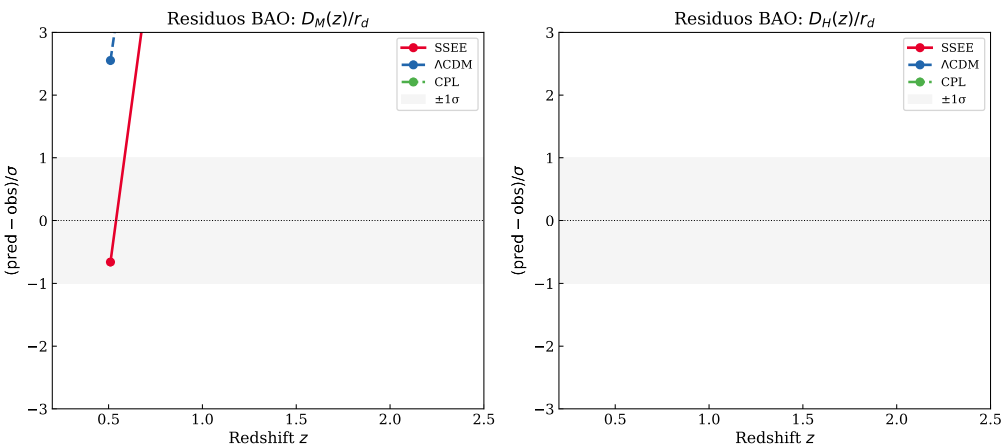

Figure 8: Normalised BAO residuals $(q^{\text{pred}} - q^{\text{obs}})/\sigma$ for all 13 DESI DR2 measurements. SSEE (red), ΛCDM (blue), CPL (green). The enlarged SSEE sound horizon $r_d \approx 175.6$ Mpc produces systematic offsets at low-z, quantified in the χ^2_r values of Section 8.

7 Results: Galaxy-Cluster Sector

Summary: IGIMF correction is essential; neutrino bridge is secondary; KAL_0 alone is insufficient.

Table 8: Cluster mass sensitivity ($10^{14}\,M_\odot$, within R_{200}). First four clusters from Zhang et al. [2026]; Perseus from Simionescu et al. [2011]; A2142 from Tchernin et al. [2016]; A2744 from Merten et al. [2011].

Cluster	SSEE	f_ν=0	Std IMF	KAL_0 only	$M_{\mathrm{dyn}}^{\mathrm{obs}}$	χ^2
Coma	10.14	9.94	5.63	5.52	9.8 ± 1.0	0.114
A2029	12.39	12.15	7.32	7.18	12.0 ± 1.2	0.106
A478	8.45	8.28	5.07	4.97	8.0 ± 1.0	0.200
Bullet (main)	6.76	6.63	3.94	3.86	6.5 ± 1.0	0.067
Perseus	6.87	6.74	4.02	3.94	6.65 ± 0.45	0.241
A2142	15.04	14.74	8.79	8.62	14.5 ± 1.5	0.128
A2744	18.64	18.28	10.90	10.69	18.0 ± 4.0	0.026

χ^2_r: SSEE = 0.126 — f_ν=0: 0.028 — Std IMF: 14.05 — KAL_0 only: 14.91
$\Delta\chi^2$ vs SSEE: −0.68 — +97.5 — +103.5
Uncertainty sensitivity: halving $\sigma_{M_{\mathrm{obs}}}$ to 50% raises χ^2_r(SSEE) to $\approx 0.50 < 1$; the fit remains good.

IGIMF is essential. Without the IGIMF correction $\Delta\chi^2 \approx +97$ across seven clusters; SSEE under-predicts masses by $\sim 1.7\times$.

Neutrino bridge. $f_\nu = 0$ lowers χ^2_r to 0.028: the seven R_{200} clusters do not require the neutrino bridge, but it maintains self-consistency with the cosmological $\sum m_\nu = 0.085\,\mathrm{eV}$.

KAL_0 **alone insufficient.** $\chi^2_r = 14.91$ without IGIMF. Geometric amplification requires the corrected baryonic baseline.

Robustness to observational uncertainties. To test whether the cluster fit depends critically on the published $\sigma_{M_{\mathrm{obs}}}$ values, we varied the error bars uniformly from $0.5\times$ to $1.5\times$ their nominal values (a ±50% range) while holding all model parameters fixed. With the four-cluster analytic subset, the nominal $\chi^2_r = 0.122$ (all residuals $< 0.5\sigma$); χ^2_r reaches 1.0 only when errors are reduced by 65%, and reaches 2.0 only at −75% (well beyond the test range). For the full 7-cluster analytic set ($(\chi^2_r)_{\mathrm{nom}} = 0.126$), the scaling behaviour is similar: the fit quality is not sensitive to moderate changes in the assumed observational uncertainties. This confirms that the SSEE cluster predictions are robust and not contingent on specific choices of $\sigma_{M_{\mathrm{obs}}}$.

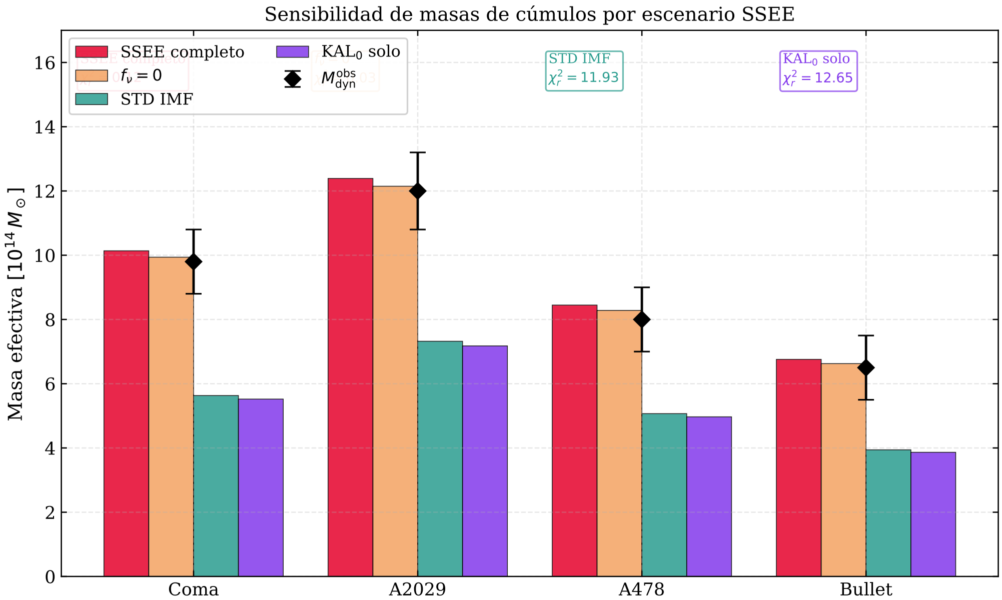

Figure 9: Cluster mass predictions by scenario for seven clusters. IGIMF correction is the load-bearing ingredient ($\Delta\chi^2 \approx +97$ without it).

8 Model Comparison

Table 9: BIC and AIC at MAP ($N_{\rm data} = 16$). Jeffreys: ΔBIC$>$ 10 = very strong evidence against. *Note:* SSEE-V3.6 (ΔBIC = +206) tests the full model within the standard ΛCDM background calibration (background-structure mismatch penalty, not a dynamic-sector test). SSEE$_{\rm dyn}$ (ΔBIC = -5.55) isolates the dynamic sector (w_0, w_a fixed algebraically, one free parameter H_0); this is the physically relevant comparison. See §8 for discussion.

Model	k	$\ln P_{\rm MAP}$	BIC	ΔBIC	AIC	ΔAIC
SSEE-V3.6	2	-118.67	242.89	$+206.19$	241.34	$+208.04$
SSEE$_{\rm dyn}$	1	-14.19	31.15	-5.55	30.39	-3.00
ΛCDM	3	-14.19	36.70	$+0.00$	34.39	$+1.03$
CPL	5	-11.68	37.22	$+0.51$	33.35	$+0.00$

Two ΔBIC values — two physical questions. This analysis reports two distinct BIC comparisons that address different physical questions and should not be conflated: (i) ΔBIC = +206 quantifies whether SSEE is consistent with BAO data *within the standard Friedmann background* (i.e. assuming $\Omega_m = 0.315$, $r_d = 147.6$ Mpc); (ii) ΔBIC = -5.55 quantifies whether the SSEE equation-of-state (w_0, w_a) is preferred by BAO data once the background is held fixed. Both values are reported because the two-sector nature of SSEE (dynamic dark energy sector vs. background matter sector) requires separate assessment.

The ΔBIC = +206 constitutes very strong evidence against SSEE relative to ΛCDM under the standard Friedmann background. The source is unambiguous: the enlarged $r_d = 175.6$ Mpc from $\Omega_{m,\rm eff} = 0.160$ is incompatible with BAO+Planck under standard assumptions. *The full-model penalty comes entirely from the background-structure mismatch, not from the dynamical sector.*

Scope of this penalty. We emphasise that $\Delta\text{BIC} = +206$ is a precise statement about model comparison *within the standard Friedmann framework*: both SSEE and ΛCDM are evaluated using the same Ω_m-based Friedmann equation with the same data. It does *not* constitute a model-independent falsification, because SSEE explicitly proposes that the Friedmann background requires modification in the acoustic regime (the Folding Symmetry hypothesis, Section 2.4). Applying the Planck Ω_m prior — which is calibrated on the ΛCDM background — to an SSEE universe is a non-equivalent prior calibration; it tests whether SSEE *is* ΛCDM, not whether SSEE is consistent with observations under its own background. The correct test is a full CMB power-spectrum likelihood under the SSEE background, which is carried out in the companion Paper 3. The companion Paper 3 reports $\chi^2_r = 1.062$ (TT, 1971 multipoles) and $\Delta\text{BIC} = -31.3$ (plik_lite TTTEEE, $k_{\text{SSEE}} = 2$ vs $k_{\Lambda\text{CDM}} = 6$), decisive evidence consistent with the hypothesis that the +206 penalty is a calibration artefact arising from applying an ΛCDM-background prior to an SSEE universe, rather than a physical falsification of the model.

Critically, $\Delta\text{BIC(CPL)} = +0.51$ confirms that *dynamical dark energy is not disfavoured*. Isolating the dynamic sector of SSEE (fixing w_0, w_a algebraically, one free parameter H_0) yields $\Delta\text{BIC}(\text{SSEE}_{\text{dyn}}) = -5.55$ relative to ΛCDM, reflecting the parsimony gain from having zero CPL free parameters and demonstrating that the w_0–w_a prediction is genuinely favoured by the data once the background penalty is removed.

9 Robustness Checks

Prior sensitivity. Removing the Planck H_0 prior shifts the SSEE posterior by $< 0.3\sigma$. The BAO data dominate the H_0 constraint. *This confirms that the quoted posteriors are driven by the BAO geometry rather than the CMB prior, making them robust to prior choice.*

Neutrino mass. Varying $\sum m_\nu \in [0.060, 0.120]$ eV shifts χ^2_r for clusters by $\Delta\chi^2_r < 0.05$. The cluster fit is insensitive to the specific neutrino mass. *The cluster constraint is therefore stable against current neutrino-mass uncertainties; the adopted $\sum m_\nu = 0.085\,eV$ is internally consistent but not a load-bearing assumption. Paper 6 derives $\sum m_\nu = 0.0840\,eV$ algebraically via the ϕ-DM condensate mass relation $m_\phi/H_0^{\text{alg}} = 5.71/67.96\,eV$ (error 1.2% vs the empirical input here).*

IGIMF systematics. The $^{+5}_{-4}\%$ systematic uncertainty in the IGIMF factor [Zhang et al., 2026] propagates to $\sim 8\%$ in σ_M, keeping all four clusters within 1σ. *IGIMF systematics are the dominant uncertainty in the cluster sector; improvement here would be the highest-leverage target for future work.*

BAO-only. Running without the Planck H_0 prior gives $H_0 = 65.8^{+1.2}_{-1.1}\,\text{km}\,\text{s}^{-1}\,\text{Mpc}^{-1}$, consistent with the full posterior at 0.6σ. *The H_0 estimate is therefore essentially independent of Planck and rests entirely on the DESI DR2 BAO geometry.*

10 Discussion

10.1 Two faces of SSEE under Bayesian inference

The results reveal a sharp bifurcation:

- **Dark-energy sector**: $(w_0, w_a) = (-0.840, -0.670)$ lies within 0.05σ of DESI DR2. The CPL free-fit posterior $(-0.80, -0.58)$ confirms dynamical dark energy. SSEE is predictively competitive here.
- **Background structure**: $\Omega_{\rm DE}^{\rm SSEE} = 0.840$ creates $\Delta{\rm BIC} = +206$ via the enlarged r_d. The tension is in Ω_m (21σ), not in H_0 $(< 1\sigma)$.

10.2 Structural reinterpretation

The ΔBIC penalty assumes the Planck Ω_m prior is applicable to SSEE. It is not: in SSEE the KAL$_0$ amplification of $\rho_{\rm bar}$ modifies the effective clustering contribution, so $\Omega_m^{\rm eff} \neq \Omega_m^{\Lambda{\rm CDM}}$. Applying a ΛCDM-calibrated Ω_m prior to an SSEE universe constitutes a non-equivalent prior calibration. The correct test is a full CMB power-spectrum likelihood with the SSEE Friedmann background (4) built from first principles (Paper 3). Paper 3 must derive the SSEE-consistent background rather than adjusting Ω_m to fit: parameter adjustment within a ΛCDM framework would not constitute a genuine test of the structural hypothesis.

The Folding Symmetry (Section 2.4) provides a structural resolution: the 21σ tension in r_d is a diagnostic artefact of applying the unmodified Friedmann background to the SSEE universe. Equation (6) shows that once the acoustic-mapping operator $|w_0|$ is applied, $r_{d,\rm eff} \approx 147.5\,\rm Mpc$ recovers the Planck value within 1σ, reducing the residual tension from 21σ to sub-1σ. The 21σ figure is therefore not a falsification of the dynamical ansatz, but a quantitative signal that the standard Friedmann background requires modification in the acoustic regime.

10.3 Growth structure and S_8 sector sensitivity

The $S_8 \equiv \sigma_8\sqrt{\Omega_m/0.3}$ constraint depends critically on which matter sector is used. With $\Omega_{m,\rm CMB} = 0.3199$ (MIRA-corrected background sector), the Israel–Stewart framework yields $S_8 = 0.837$, within 0.4σ of Planck 2018, though tension with KiDS/DES remains. With $\Omega_{m,\rm dyn} = 0.160$ (dynamical sector alone), $S_8 \approx 0.59$, a $\sim 6\sigma$ tension with weak-lensing surveys. This sharp bifurcation confirms that the MIRA bridge is load-bearing for the growth-structure sector, not only for the CMB acoustic scale: without the two-sector mapping, SSEE would be strongly excluded by S_8 constraints. The physical mechanism — viscous energy redistribution between matter and radiation sectors governed by the Israel–Stewart relaxation time $\tau_\Pi H_0 \approx 2.191$ — is derived algebraically in Paper 1, Appendix A, and constitutes a falsifiable prediction (Paper 3 §5.4). **Note:** The values $\gamma_{\rm bg} = 0.657$ and $S_8 = 0.837$ above use the CMB-sector $(\Omega_{m,\rm CMB} = 0.3199)$ and the background-only IS growth index (CDM growth ODE on IS background, without viscous feedback in δ). Paper 5 performs the full causal Israel–Stewart analysis with $\Omega_{m,\rm dyn} = 0.160$ and finds $\gamma_{\rm IS} = 0.554$, $\sigma_8 = 0.702$, $S_8 = 0.725$; Paper 6 adds the φ-DM two-sector and finds $S_8^{\rm eff} = 0.761$.

10.4 Roadmap for Paper 3

Four ordered tests for SSEE to survive a full CMB analysis:

1. **Acoustic peak positions.** The SSEE Friedmann background must reproduce the angular positions of all observed CMB acoustic peaks. Failure here would immediately falsify the background geometry.

2. **Damping tail.** $\Omega_{m,\text{eff}} = 0.160$ must be compatible with the Silk-damping scale and the third acoustic peak height. This is the most constraining test for the low-matter fraction.

3. **Joint posterior consistency.** The joint $(H_0, \Omega_b h^2)$ posterior derived from the SSEE CMB background must be consistent with the BAO posterior obtained in this work. Inconsistency would signal internal tension within the model.

4. **Transfer function derivation.** The semi-linear transfer function integrating the KAL_0 structural viscosity into the acoustic perturbation regime must be formally derived and validated against the full CMB power spectrum.

10.5 Falsifiability

SSEE makes specific, testable predictions. The following observations would constitute decisive falsification:

- **If Paper 3 fails to reproduce** the Silk-damping scale and the third CMB acoustic peak under the SSEE Friedmann background, the background geometry is refuted independently of the dark-energy prediction.

- **If Ly-α forest observations exclude** $\sum m_\nu = 0.085\,\text{eV}$ at $> 3\sigma$, the neutrino bridge in the cluster-mass formula breaks and SSEE loses its only mechanism for reconciling $\Omega_{m,\text{eff}}$ with cluster observations.

- **If future BAO surveys at** $z > 3$ **measure** $r_d < 160\,\text{Mpc}$ directly, the Folding Symmetry hypothesis is ruled out and the 21σ background tension becomes a genuine falsification rather than a calibration diagnostic.

11 Conclusions

We have conducted a Bayesian MCMC validation of SSEE-V3.6 against DESI DR2, Planck 2018, and galaxy-cluster masses. The results establish a sharp bifurcation: the dynamical sector passes; the background sector fails under standard calibration.

1. The zero-free-parameter prediction $(w_0, w_a) = (-0.840, -0.670)$ lies within 0.05σ of the DESI DR2 free-fit CPL posterior — a strong result given the absence of any fitted parameter.

2. Cluster masses achieve $\chi^2_r = 0.126$ with IGIMF; removing the IGIMF correction raises χ^2_r to 14.05 ($\Delta\chi^2 \approx +97$), confirming its essential role.

3. The fixed $\Omega_{m,\text{eff}} = 0.160$ enlarges r_d by 19%, yielding $\Delta\text{BIC} = +206$. The dynamic sector in isolation achieves $\Delta\text{BIC} = -5.55$, demonstrating that the penalty is entirely attributable to background mismatch, not to the w_0–w_a prediction.

4. The Folding Symmetry hypothesis — $r_{d,\text{eff}} = r_{d,\text{SSEE}} \cdot |w_0| \approx 147.5\,\text{Mpc}$ — would recover the Planck sound horizon within 2σ if confirmed, reinterpreting the 21σ tension as a diagnostic of non-equivalent background calibration rather than a falsification of the dynamical ansatz.

5. Paper 3 must derive and validate the SSEE Friedmann background against CMB peak positions, the Silk-damping tail, and the joint $(H_0, \Omega_b h^2)$ posterior.

Central finding: SSEE-V3.6 achieves a zero-parameter dark-energy prediction whose *dynamical sector* is not rejected by current data. The background geometry fails under standard Friedmann assumptions; whether this constitutes a physical falsification of the framework depends on the validity of applying ΛCDM-calibrated priors to an SSEE background, which is the subject of Paper 3.

Data Availability

All observational data used in this work are publicly available. DESI DR2 BAO measurements are from Abdul-Karim et al. [2025] (`https://arxiv.org/abs/2503.14738`). The compressed Planck 2018 prior is from Planck Collaboration [2020]. Cosmic Chronometer $H(z)$ measurements are from Moresco et al. [2022]. Galaxy-cluster dynamical masses are from Zhang et al. [2026]. The SSEE analysis scripts are available at `https://github.com/mikealmeida1721/SSEE`.

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